SESSION 1: INTRODUCTION TO R, R-STUDIO, AND DATA FRAMES

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- R and RStudio
- Data types
- Data frames

1. R & RSTUDIO

What is R?

What is RStudio?

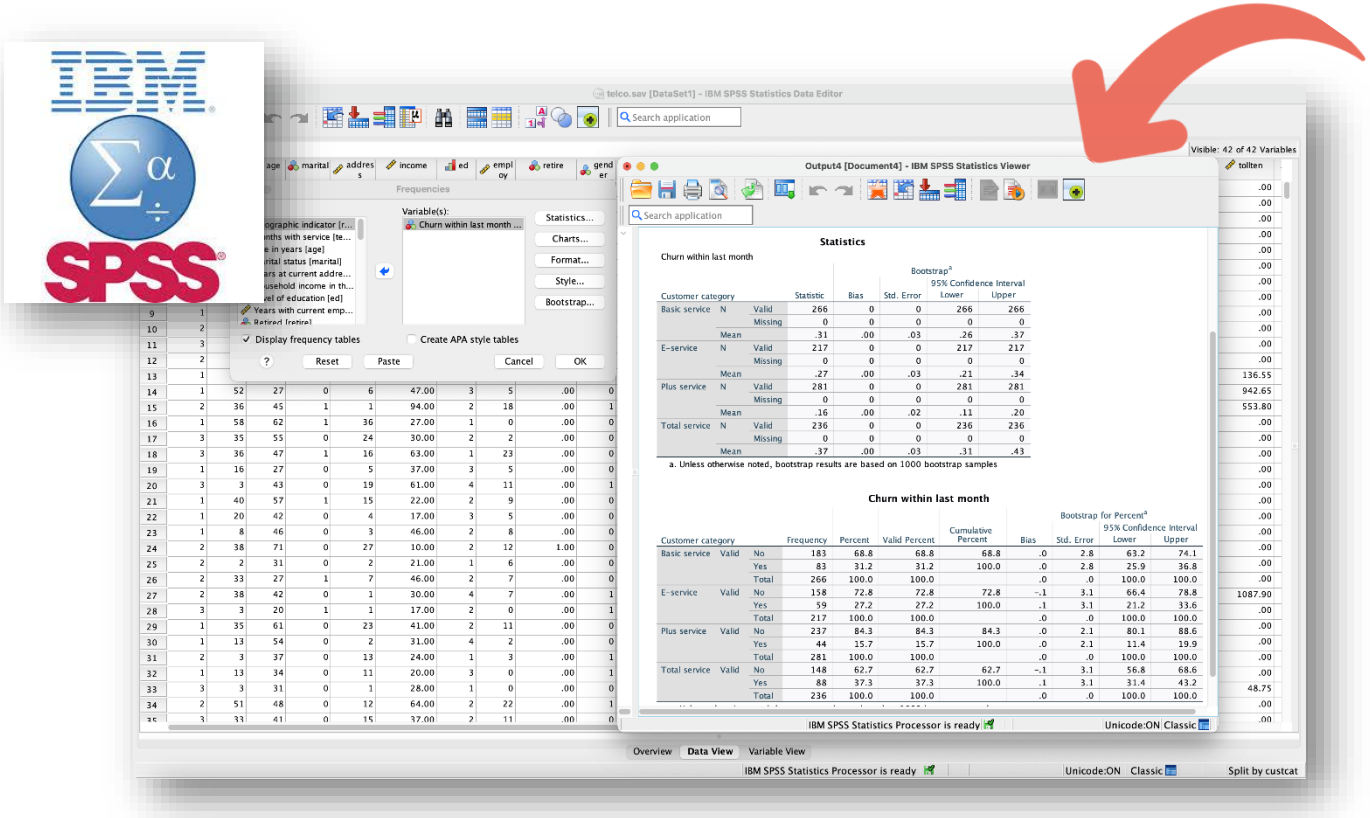
APPLICATION SOFTWARE

The main function of application software is to help and assist a user to perform specific tasks in a computer. Application programs can be developed with different software languages and tools.

PROGRAMMING LANGUAGES

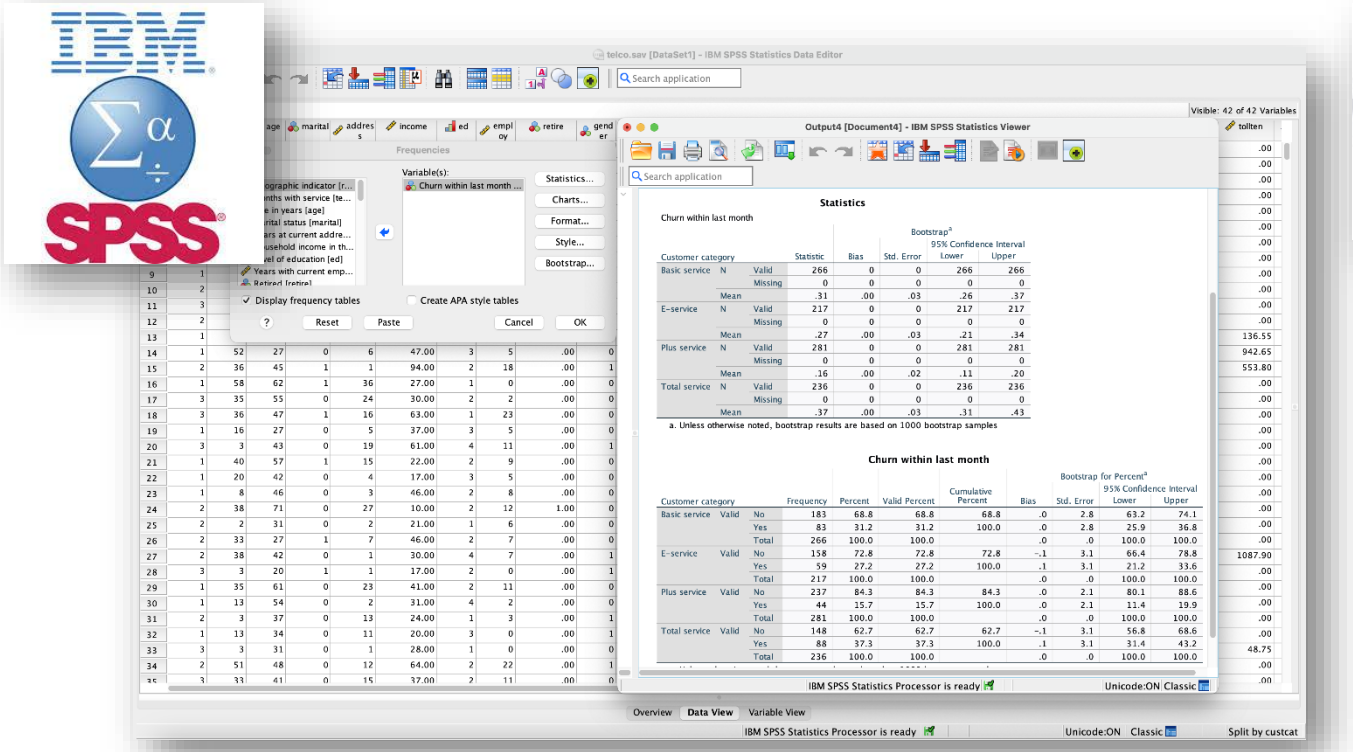
Programming languages are used to write programs that allow user/machine communication.

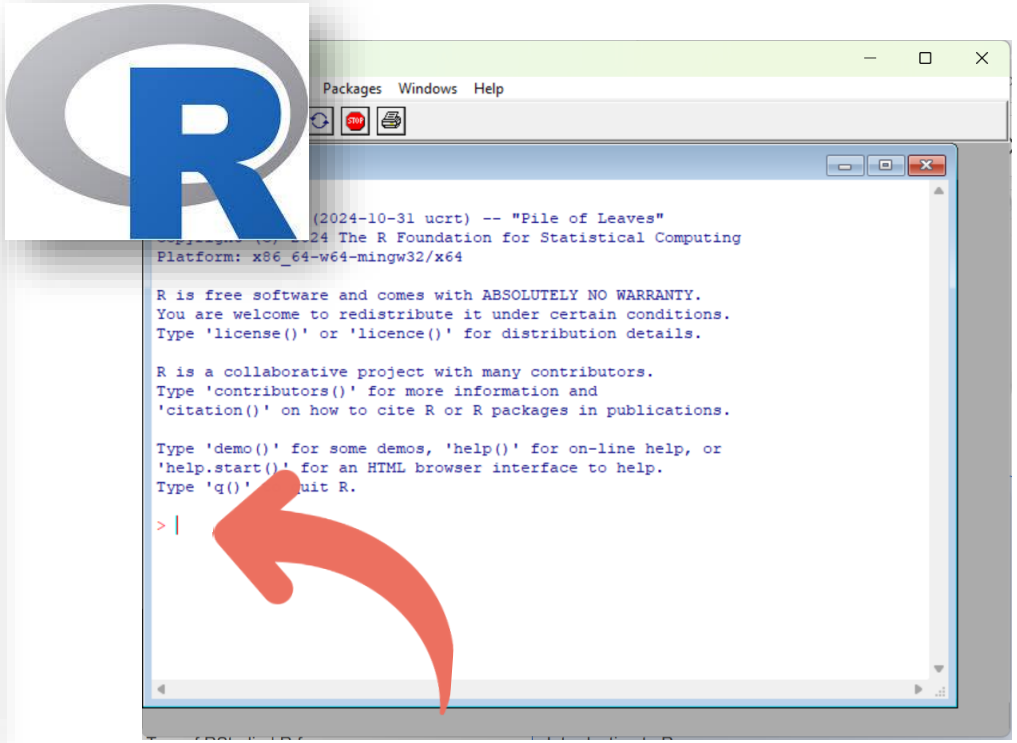
APPLICATION SOFTWARE VS. PROGRAMMING LANGUAGE



If you want to do something, then you just need to "click click click"

APPLICATION SOFTWARE VS. PROGRAMMING LANGUAGE





If you want to do something, then you just need to write code, “code code code”

THE LOGIC OF R

THE LOGIC OF R

- Data is stored in variables.
  -
  -
-

```
> 2
[1] 2
> "hello"
[1] "hello"
```

THE LOGIC OF R

- Data is stored in variables.
  - This means that we always need to assign numbers, strings, and databases to variables.
  - A variable is a string of characters. It is assigned a value by using the special characters “=” or “<-”.

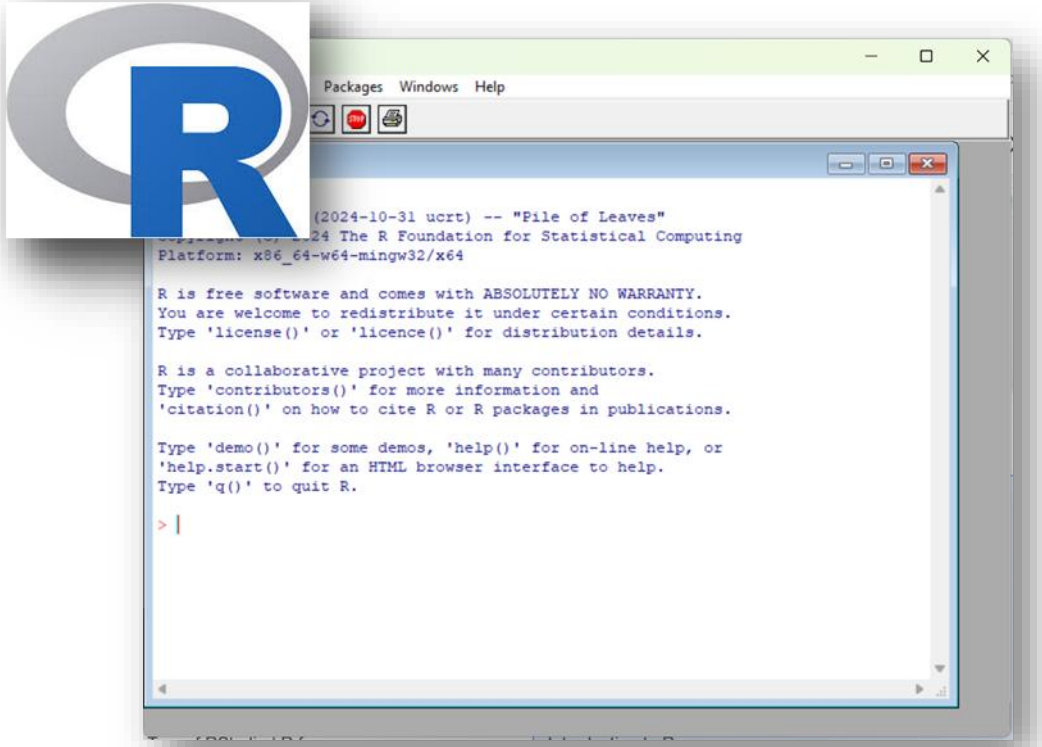
```
> 2
[1] 2
> "hello"
[1] "hello"
>
> x=2
> y<-"hello"
>
```

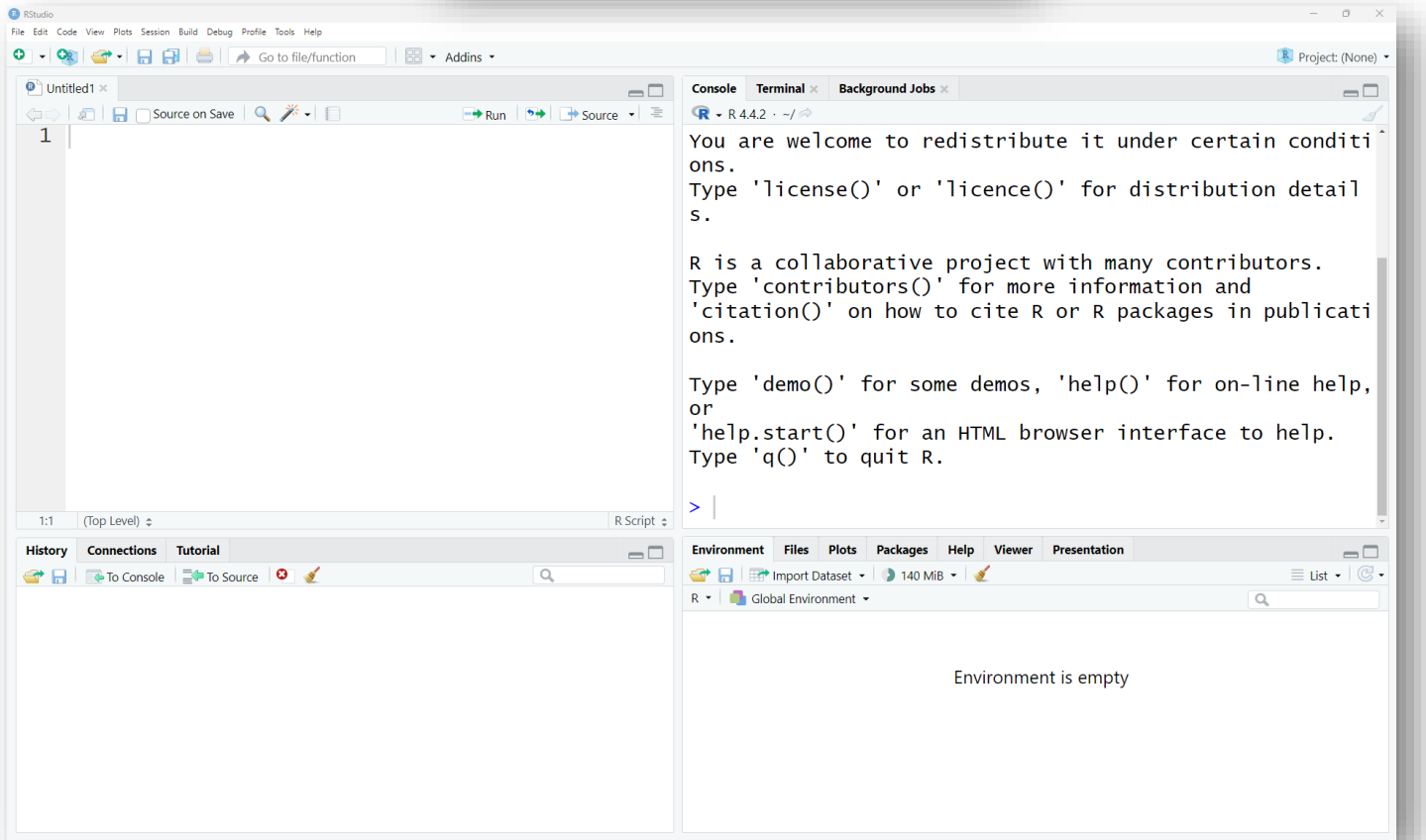
THE LOGIC OF R

- Data is stored in variables.
  - This means that we always need to assign numbers, strings, and databases to variables.
  - A variable is a string of characters. It is assigned a value by using the special characters “=” or “<-”.
- Variables are case sensitive, i.e. “HELLO” is not the same as “hello” or “Hello”. So, always try to remember the exact characters you used.

```
> 2
[1] 2
> "hello"
[1] "hello"
>
> x=2
> y<-"hello"
>
> HELLO
Error: object 'HELLO' not found
>
> HELLO="hello"
> hello<-"no way"
>
> HELLO
[1] "hello"
>
> hello
[1] "no way"
>
```

IBM SPSS + R = RStudio®





Now you can "click click click" to access certain functions, but you still need to "code code code" to tell the computer what to do.

LET'S OPEN RSTUDIO!

2. A SHORT INTRODUCTION TO R DATA TYPES

Primitive Data Types
Vectors

WHY ARE DATA TYPES RELEVANT?

This is a dataframe

```
                  mpg cyl disp  hp drat   wt ...
Mazda RX4        21.0   6  160 110 3.90 2.62 ...
Mazda RX4 Wag    21.0   6  160 110 3.90 2.88 ...
Datsun 710       22.8   4  108  93 3.85 2.32 ...
                 .............
```

It is form by numbers and strings.

```
                  mpg cyl disp  hp drat   wt ...
Mazda RX4        21.0   6  160 110 3.90 2.62 ...
Mazda RX4 Wag    21.0   6  160 110 3.90 2.88 ...
Datsun 710       22.8   4  108  93 3.85 2.32 ...
                 .............
```

These numbers and text are stored in something called a vector.

```
                 mpg cyl disp  hp drat   wt ...
Mazda RX4       21.0   6  160 110 3.90 2.62 ...
Mazda RX4 Wag   21.0   6  160 110 3.90 2.88 ...
Datsun 710      22.8   4  108  93 3.85 2.32 ...
                 .............
```

Data Types

A dataframe is the result of joining vectors that contain numbers, strings, or logical values.

Float/ Double

Integers

Character & String

Logical

Data Frame

```
> mtcars
                mpg cyl disp  hp drat   wt ...
Mazda RX4      21.0   6  160 110 3.90 2.62 ...
Mazda RX4 Wag  21.0   6  160 110 3.90 2.88 ...
Datsun 710     22.8   4  108  93 3.85 2.32 ...
                ............
```

Vector

❖ DATA TYPES

A data type is a set of values and operations associated with those values.

Primitive: These are the simplest types of data, because they are not constructed from other types and are unique entities that cannot be decomposed into others. These data types are defined by a set of values and a set of operations that act on those values.

- Integer
- Floating/Double
- Character and String
- Logical

PRIMITIVE DATA

Integer	Values	-inf, …, -2, -1, 0, 1, 2, …, inf
	Operations	*, +, -, %, /, …
Float	Values	…, -0.6, 0.0, 0.5, …
	Operations	*, +, -, %, /, …
Character/ String	Values	‘\n’, ‘A’, …, ‘Z’, ‘a’, …, ‘z’
	Operations	<, >, ==, …
Logical	Values	TRUE, FALSE
	Operations	<, >, ==, …

- **Numerical Type:**
  - Integer
  - Floating/Double
- **Character:** This type of data consists of the set of characters available for a specific language on a specific computer. The most used character code is ASCII.
  - A **String** is multiple characters together
- **Logical:** This is the simplest data type, as it only has two values (TRUE (1) and FALSE (0))

PRIMITIVE DATA

Integer	Values	-inf, …, -2, -1, 0, 1, 2, …, inf
	Operations	*, +, -, %, /, …
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	Operations	*, +, -, %, /, …
Character/ String	Values	‘\n’, ‘A’, …, ‘Z’, ‘a’, …, ‘z’
	Operations	<, >, ==, …
Boolean	Values	TRUE, FALSE
	Operations	<, >, ==, …

- **Numerical Type:**
  - Integer: Age
  - Floating/Double: Height
- **Character:** This type of data consists of the set of characters available for a specific language on a specific computer. The most used character code is ASCII.
  - A **String** is multiple characters together: Country names
- **Logical:** This is the simplest data type, as it only has two values (TRUE (1) and FALSE (0)): Gender – Woman (1), Man (0)

EXERCISE 1.1

Operators					
Aritmetic		**Comparative**		**Logical**	
+	addition	<	less than	! x	logic NO
-	subtraction	>	more than	x & y	element-wise AND
*	multiplication	<=	less or equal than	x && y	single comparison AND
/	division	>=	more or equal than	x | y	element-wise OR
^	power	==	equal than	x || y	single comparison OR
%/%	integer division	!=	different than	xor(x,y)	exclusive OR

Using the operators shown above and the variables

```
A<-5
B<-10
```

```
C<- -10
D<- -8
```

Try to get the following results:

- 25
- TRUE
- 3

❖ DATA TYPES

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- Integer
- Floating/Double
- Character and String
- Boolean

Composite Data: These are types of data whose values are collections of primitive data.

- Vectors

VECTORS

A vector is a collection of observations or measurements of the same type, for example the ages of 50 people or the number of coffees we drink on different days of the week. Vectors are the simplest types of aggregated data; it is with them that more complex structures are created.

In R, the function with which a vector is created is **c()**:

```
myvec <- c(1,3,1,42)
myvec
```

2. DATAFRAMES

Dataframes.
From table/csv to dataframes & vice-versa.
Tidyverse.

❖ DATAFRAMES

A dataframe is a special type of R object where all the items are vectors of the same length. In a dataframe, the elements of different columns can be of different types.

In the example beneath, we are constructing a dataframe with 3 columns (‘n’, ‘s’, and ‘b’). ‘n’ is numeric. ‘s’ is string. ‘b’ is Boolean.

To glue them together, we use the R function **data.frame()**.

Data Frame Construction

```
n = c(2, 3, 5)
s = c("aa", "bb", "cc")
b = c(TRUE, FALSE, TRUE)
df = data.frame(n, s, b)
```

❖ DATAFRAMES

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Data Frame Construction

```
> n = c(2, 3, 5)
> s = c("aa", "bb", "cc")
> b = c(TRUE, FALSE, TRUE)
> df = data.frame(n, s, b)
```

Exercise 2.1: Follow the code to create the “df” dataframe.

OPEN AND SAVE

With the *foreign* library you can manipulate databases other than ".csv"

Open: read.csv()

You can open "csv" files using this function.

```
read.csv(file, header = TRUE, sep = ",", quote = "\"",
         dec = ".", fill = TRUE, comment.char = "", ...)
```

Save: write.csv()

```
write.table(x, file = "", append = FALSE, quote = TRUE, sep = " ",
            eol = "\n", na = "NA", dec = ".", row.names = TRUE,
            col.names = TRUE, qmethod = c("escape", "double"),
            fileEncoding = "")

write.csv(...)
```

EXERCISE 2.2

Save the “df” dataframe. Do you need the row.names parameter?

```
DIR="WRITE HERE THE PATH TO THE DATA\\"
write.csv(Artist ,paste0(DIR,"Data_Name.csv"),row.names = FALSE)
```

With the command **help(“paste0”)** you can check what this function do.

REFERENCES

- Albert, Jim, and Maria Rizzo. *R by Example: Concepts to Code*. Use R! New York, NY: Springer New York, 2012. https://doi.org/10.1007/978-1-4614-1365-3.
- Davies, Tilman M. *The Book of R: A First Course in Programming and Statistics*. San Francisco: No Starch Press, 2016. https://web.itu.edu.tr/~tokerem/The_Book_of_R.pdf
- Wickham, Hadley, Mine Çetinkaya-Rundel, and Garrett Grolemund. *R for data science*. " O'Reilly Media, Inc.", 2023. Accessed May 7, 2024. https://r4ds.hadley.nz/.

❖ SESSION 2: TIDYVERSE HTTPS://WWW.TIDYVERSE.ORG/

R packages for data science

The tidyverse is an opinionated **collection of R packages** designed for data science. All packages share an underlying design philosophy, grammar, and data structures.

Install the complete tidyverse with:

```
install.packages("tidyverse")
```

Join us!

https://forms.office.com/e/8Bgd2YsasJ

All about the lab:

https://societal-analytics.nl/

Contact us at:

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https://sofiag1l.github.io/

THANKS!

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